DETAIL SPECIFICATION SHEET

ARC FAULT PROTECTION SYSTEM, PHOTOSENSOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32656.

REQUIREMENTS:

Photosensor design. The photosensor shall consist of a photodiode element, an operational amplifier, and a built-in-test (BIT) circuit in a shielded plastic case. A photodiode shall convert the incident radiant energy into a proportional electrical current that is amplified and converted into an equivalent voltage by an integral amplifier. The BIT circuit shall consist of light emitting diodes (LED) and an appropriate circuit that shall exercise the photodiode when stimulated by an outside test command. The amplifiers in the photosensor are set to produce a voltage within the operating range of the design specification. When the photosensor is directly exposed to ambient compartment lighting, the combination of selective coatings on the lens and the gain settings keep the photosensor signal significantly below the alarm threshold while an arc will cause the output to be at least ⅔ of the full-scale voltage output.

BIT. BIT excitation voltage and current shall be compatible with the vendor's control unit. The BIT response time of the sensor shall be less than 1.0 millisecond. The BIT LED output shall saturate the output of the detector at 0, 25, and 75 °C.

Sensitivity. The photosensor assembly shall have a nominal sensitivity of 46,000 volts per watt ±10 percent to 633±1 nanometers light on the sensor face to the amplifier output.

Envelope. The photosensor, without respect to connecting hardware and cabling, shall occupy an envelope of no more than 5 centimeters (2 inches) wide by 5 centimeters (2 inches) high by 2 centimeters (0.75 inch) diameter.

Photodiode assembly. The photodiode shall have a minimum active area of 5.1 square millimeters when integrated into the assembly. The sensitivity of the photodiode used shall be greater than 0.4 amperes per watt at 800 nanometers and 0.3 amperes per watt at 632 nanometers.

Viewing angle. The photosensor window and housing shall permit full illumination of the detector with a viewing angle of 45 degrees from the normal to the photodiode surface in all directions (90-degree cone) minimum.

Optical window. An ultraviolet-grade glass window with a transmission of 50 percent at 220 nanometers and 90 percent at 300 nanometers shall be used for the photodiode assembly.

Transmission/reflectivity. The photodiode assembly shall have an anti-reflective coating outside the surface of the window to achieve reflectivity greater than or equal to 99 percent at 880 nanometers, average reflectivity greater than or equal to 95 percent from 860 to 915 nanometers, and average transmission greater than or equal to 85 percent from 300 to 825 nanometers.

Case shielding. The photosensor enclosure shall be shielded from allowing the transmission of electromagnetic radiation through the enclosure with the exception of the lens, which shall allow transmission of visible and ultraviolet light in accordance with the optical window requirement.

Power. The power required for the photosensor shall be set to the designed operating voltage and shall be supplied from outside the device.

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Background lighting. The photosensor shall be furnished with a means to block directional light, such as that from a flashlight. This 'light shield' shall be removable and able to block directional light in any minimum 45-degree arc, along the plane normal to the view of the sensor. The light shield shall be capable of installation to block background light along any 45-degree arc at the installer's discretion.

EXAMINATIONS AND TESTS:

LED and detector output test. All photosensors shall be tested to ensure all LEDs are operational and the detector output is greater than 100 microamps during the temperature ranges of 0, 25, and 75 °C. The photodiode cathode and anode shall be connected to a picoammeter. The LED anode shall be connected to a 360-ohm resistor and then to the plus output of a direct current (DC) power supply. The LED cathode shall be connected to the return of the power supply. The devices shall be tested at 25 °C for voltage drop across the current limiting resistor and photodiode current due to LED emittance. All LEDs shall be observed visually to ensure that they emit approximately the same amount of light. The device shall then be tested at 0 °C and then at 75 °C. Acceptable units shall exhibit a voltage drop across the resistor of less than 7.0 volts and the photo detector shall have an output of greater than 100 micro amps (µA) at all three temperatures.

Photodiode functional test. All photosensors shall be tested to ensure that the photodiode is operating properly by measuring various parameters. The parameters shall be measured with appropriate test equipment at an ambient temperature of 25 °C and with the photodiode in the dark. The parameters that shall be measured are dark current, shunt resistance, and series resistance. The dark current shall be less than 10 nano-amperes at 1-volt reverse bias, the shunt resistance shall be greater than 50 megohms at ±10 millivolts, and the series resistance shall be less than 1,000 ohms at 0.75 volt-forward bias.

Final detector responsivity test. Every photosensor shall be tested to measure the final responsivity of the device with its detector and amplifier while utilizing a 633±1-nanometer light source and the BIT. The appropriate test equipment and test fixtures shall be connected to a reference detector calibrated by the National Institute of Standards and Technology (NIST) with known responsivity at 633±1 nanometers. The designed operating voltage shall be supplied to the detector and an adjustable DC power supply shall be supplied for the BIT. In a darkened room, the output readings for the calibrated reference detector shall be noted and recorded. Using a Lumentum Helium-Neon laser, model 1508-0 (633±1 nanometer), or equivalent, the laser shall be turned on and placed at a distance that does not saturate the detector but produces a detector output which is at least ten times greater than the reading taken in the darkened room without the laser. The output reading for the reference detector shall be noted and recorded, and the distance of the reference detector from the laser shall be recorded. The reference detector shall be replaced with the sensor under test. The sensor shall be covered with black felt cloth to remove the light source from the detector. The sensor power supply shall be turned on; the offset voltage shall be noted and recorded. The cover shall be removed, the detector shall be moved away from the light source the same distance previously recorded and under the same lighting conditions, and the output voltage from the sensor shall be noted and recorded. The responsivity of 633±1 nanometers for the sensor under testing shall be calculated by using the following formula:

$$R_{\lambda}\,(DUT) = [V_{out}(DUT)][R_{\lambda}(REF)]/\,[V_{out}(REF)]$$

V_{out} (DUT) is the voltage measured when the cover is removed from the sensor, R_{λ}(REF) is the known responsivity of the reference detector at 633±1 nanometers, and V_{out} (REF) is the voltage measured from the reference detector with the laser on.

Every photosensor shall have a responsivity of 46,000 volts per watt ±10 percent when using a 633±1-nanometer source.

With the photosensor covered, the BIT voltage shall be gradually increased from 0 to that required for the sensor to produce an output greater than ⅔ the full-scale voltage. The BIT voltage shall be recorded. The detector output shall reach a value greater than ⅔ the full-scale output voltage with a BIT voltage less than the BIT operating voltage at 25 °C.

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Operating temperature range test. The photosensor shall be connected so that the designed operating voltage is supplied to the sensor and an adjustable DC power supply shall be supplied for the BIT. The sensor shall be placed in an environmental chamber and the temperature shall be set to 0 °C. The BIT shall be activated with the operating voltage DC, and the LED current shall be recorded. This test shall be repeated at temperatures of +25 and +75 °C. Every photosensor tested shall cause the analog output to saturate when the BIT is activated at 0, +25, and +75 °C.

CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH
Air Force – 85
DLA – GS

Review activities:
Army – CR, MI
Navy – AS, CG, YD
Air Force – 03, 19, 84
DLA – GS8

Civil agency:
GSA – FAS

Preparing activity:
Navy – SH
(Project 5975-2019-002)

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